EMPIRICAL ROUGHNESS IN EQUITY VOLATILITY

Hurst Scaling, Realized-Variance Estimation, and Short-Maturity Option Skew

Brownian Benchmarks and Rough Stochastic Volatility Dynamics

Francesco Cammarota

ST. GEORGE CAPITAL | QUANTITATIVE RESEARCH

Abstract

This paper investigates the empirical and structural implications of modelling equity-market volatility as a rough stochastic process. Volatility is formulated through Gaussian Volterra dynamics with fractional kernels and Hurst index $H < 1/2$. Using 4,189 daily S&P 500 observations from the Oxford-Man Realized Library over 3 January 2000-6 October 2016, the paper independently estimates roughness from five-minute realized variance. A baseline structure-function specification yields $H = 0.1318$ ($R^2 = 0.99998$), with a dependence-preserving moment-block bootstrap 95% interval of [0.1077, 0.1545]. The estimate remains far below the Brownian benchmark across subsampled five-minute variance, ten-minute variance, and realized-kernel measures, although economically meaningful regime variation is present. The empirical estimate is then linked to the short-maturity skew mechanism and to controlled rough-Bergomi and Heston experiments. Evidence supports low local regularity as a robust feature of measured volatility while also showing that the magnitude of estimated roughness depends on measurement choice and market regime.

Keywords: rough volatility; fractional Brownian motion; Volterra processes; Hurst exponent; realized volatility; implied-volatility skew; rough Bergomi; stochastic volatility.

1. Introduction

Continuous-time asset-pricing models typically represent log prices as semimartingales driven by Brownian innovations, with volatility regulating the local scale of returns. Therefore, the central modelling problem is not concerned with the introduction of randomness per se, but specifying the volatility process. Constant or deterministic volatility delivers analytical tractability, while local- and stochastic-volatility extensions introduce state dependence and additional sources of uncertainty. Yet a substantial empirical literature indicates that the regularity of volatility itself is materially lower than that implied by classical Brownian diffusions.

Gatheral, Jaisson and Rosenbaum (2018) re-examine volatility smoothness using high-frequency data and report that log-volatility is well approximated, over economically relevant horizons, by a fractional process with Hurst index of order $H \approx 0.1$. Since Brownian sample paths correspond to the benchmark $H = 1/2$, these estimates imply markedly rougher local behaviour. The implication is structural: if volatility is rough, then replacing constant volatility by a conventional Markov diffusion does not merely miss a parameter value, but imposes an inappropriate local regularity class on the latent state variable.

The option market provides a complementary identification channel. Short-dated equity-index options exhibit a steep at-the-money (ATM) implied-volatility skew. Rough stochastic-volatility models link the asymptotic behaviour of this skew to the Hurst index, generating a non-trivial power law at short maturities. Bayer, Friz and Gatheral (2016) show that a parsimonious rough-volatility specification, the rough Bergomi model, can reproduce salient features of the SPX volatility surface while retaining an explicit forward-variance input. This provides a natural bridge between historical volatility dynamics and risk-neutral option prices.

This study aims to examine that bridge in a unified empirical framework. First, we formalize Brownian motion as the diffusion benchmark and embed it into the broader class of Gaussian Volterra processes. Second, we replace the regular Brownian kernel with a singular fractional kernel and characterize the role of H in covariance, increment dependence, and path regularity. Third, we estimate the scaling exponent of log-realized-volatility increments from high-frequency equity data rather than imposing H ex ante. Finally, the resulting estimate is carried into a rough Bergomi pricing experiment and compared with a conventional stochastic-volatility benchmark, while market-implied calibration is left as an empirical extension.

The contribution is deliberately empirical rather than purely expository. The analysis distinguishes three questions that are often conflated: whether observed volatility is locally rough; whether the estimated degree of roughness is stable across measurement choices and regimes; and whether that roughness improves the cross-sectional representation of option prices. The paper therefore treats the Hurst exponent as an estimand subject to statistical

© 2026 St. George Capital (SGC). All rights reserved

uncertainty, not as a calibration constant inherited from the literature. This distinction is central to the interpretation of any subsequent pricing result.

The remainder of the paper is structured as follows. Section 2 establishes the diffusion and Volterra benchmark. Section 3 introduces fractional Brownian motion and the roughness parameter. Section 4 develops the empirical identification strategy based on realized volatility and scaling laws. Section 5 turns to option-implied evidence and short-maturity skew. Section 6 specifies the rough Bergomi model and calibration architecture. Section 7 sets out the comparative pricing and robustness design; the final section discusses the economic interpretation and limitations of the evidence.

2. Diffusion Benchmark and Volterra Representation

2.1 Brownian diffusion as the local benchmark

Let $(\Omega, F, (F_t)_{t\geq 0}, P)$ be a filtered probability space satisfying the usual conditions, and let $W = (W_t)_{\geq 0}$ denote a standard Brownian motion. Thus $W_0 = 0$ almost surely, W has continuous paths, and for $0 \leq s < t$ the increment $W_t - W_s$ is independent of F_s and distributed $N(0, t-s)$. Under the geometric diffusion benchmark,

$$\frac{dS_t}{S_t} = \mu_t\, dt + \sigma_t\, dW_t. \quad (1)$$

For constant σ and μ, Itô's formula yields

$$\log S_t = \log S_0 + (\mu - \sigma^2/2)t + \sigma W_t.$$

The role of Brownian motion here is local: quadratic variation accumulates linearly in calendar time, $[W]_t = t$, and the diffusion coefficient controls instantaneous return variance. The rough-volatility hypothesis does not reject Brownian shocks to returns; it changes the regularity and dependence structure of the volatility state that scales those shocks.

2.2 Gaussian Volterra processes

A convenient class of non-Markovian Gaussian processes is obtained by convolving Brownian noise with a deterministic Volterra kernel. For $t \geq 0$, define

$$X_t = \int_0^t K(t,s)\, dW_s, \quad (2)$$

where $K(t,s)=0$ for $s>t$ and $K(t,\cdot)$ is square-integrable on $[0,t]$. The kernel determines the covariance operator and hence the temporal dependence and local regularity of X. By Itô isometry, for $s,t \geq 0$,

$$\mathrm{Cov}(X_t, X_s) = E[X_t X_s] = \int_0^{\min(s,t)} K(t,u)K(s,u)\, du. \quad (3)$$

The standard Brownian case is recovered from $K(t,s)=1$. More generally, a kernel that becomes singular as s approaches t places disproportionate weight on recent innovations and generates substantially rougher paths. This kernel-based formulation is therefore the natural point of departure for fractional volatility models.

3. Fractional Dynamics and the Hurst Index

3.1 Fractional Brownian motion

Fractional Brownian motion $B^H = (B_t^H)_t \geq 0$, $H \in (0,1)$, is the centered Gaussian process with covariance

$$E[B_t^H B_s^H] = \frac{1}{2}(t^{2H} + s^{2H} - |t-s|^{2H}). \quad (4)$$

It is H-self-similar and has stationary increments. In particular, B_t^H is equal in finite-dimensional distribution to $c^H B_t^H$. For $H=1/2$, the covariance reduces to $\min(s,t)$, recovering standard Brownian motion. For $H \neq 1/2$, increments are no longer independent. The sign of the covariance between adjacent increments is negative for $H<1/2$ and positive for $H>1/2$, providing a precise distinction between anti-persistent and persistent fractional dynamics.

Path regularity is equally important. Fractional Brownian paths are locally Hölder continuous of every order strictly below H, and of no higher order. Hence $H<1/2$ corresponds to paths rougher than Brownian motion. In rough-volatility applications, the empirically relevant range is substantially below one half; the canonical evidence of Gatheral, Jaisson and Rosenbaum (2018) places H around 0.1 for log-volatility.

© 2026 St. George Capital (SGC). All rights reserved

3.2 Fractional Volterra kernel

For numerical work it is convenient to employ a one-sided fractional Volterra process. A Riemann-Liouville-type representation takes the form

$$Y_t = \sqrt{(2H)} \int_0^t (t-s)^{H-\frac{1}{2}} \, dW_s , \quad (5)$$

where the normalization is chosen so that $Var(Y_t)=t^{2H}$. When $H<1/2$, the exponent $H-1/2$ is negative and the kernel is singular at the diagonal. Recent Brownian shocks consequently receive large local weight. This singularity is the mathematical mechanism through which roughness enters the volatility state in the rough Bergomi construction.

4. Empirical Identification of Volatility Roughness

4.1 Realized-volatility proxy

The baseline empirical implementation uses the S&P 500 series from the former Oxford-Man Institute Realized Library, accessed through an archival public research mirror. The source file provides daily realized measures computed from high-frequency index observations, including five-minute realized variance, realized kernel, subsampled realized variance, bipower variation and realized semivariance. The baseline roughness series is constructed from the five-minute realized-variance measure. For trading day t partitioned into M intraday intervals, let $r_{t,j}$ denote the j-th intraday log return. The corresponding realized-variance estimator is $RV_t = \Sigma_{j=1}^{M} r^2_{t,j}$.

$$RV_t = \sum_{j=1}^{M} r_{t,j}^2 \qquad \hat{\sigma}_t = \sqrt{RV_t}, \qquad x_t = \log \hat{\sigma}_t \quad (6)$$

The data begin on 3 January 2000 and contain both conventional and noise-robust realized measures. This distinction is useful for identification because finer sampling increases the nominal information set while also increasing sensitivity to market microstructure effects. The five-minute realized variance is used as the transparent baseline; robustness is assessed with the five-minute subsampled estimator, ten-minute realized variance and the realized kernel. The empirical analysis therefore separates a baseline roughness estimate from measurement-specific perturbations rather than treating a single realized-volatility proxy as error-free.

4.2 Structure functions and scaling exponents

If x_t behaves locally as a fractional process with Hurst index H, its q-th absolute increment moment obeys a power law in the lag Δ. Define the empirical structure function

$$m(q,\Delta) = \left(\frac{1}{N_\Delta}\right) \Sigma |x_{t+\Delta} - x_t|^q . \quad (7)$$

For a monofractal fractional specification,

$$m(q,\Delta) \asymp C^{\varphi} \Delta^{\{\zeta^{\varphi}\}}, \quad \zeta^{\varphi} = qH . \quad (8)$$

Taking logarithms converts the scaling relation into a sequence of log-log regressions,

$$\log m(q,\Delta) = \alpha_q^{\varphi} + \zeta_q^{\varphi} \log \Delta + \varepsilon^{\varphi}_{q,\Delta} . \quad (9)$$

The first-stage slope ζ^{φ} is estimated separately across q. The second-stage restriction $\zeta^{\varphi}=qH$ then identifies H from the slope of ζ^{φ} on q. The empirical object of interest is not simply whether $\hat{H}<1/2$, but whether the linear scaling restriction is stable across moment orders, lag ranges, sampling frequencies and market regimes. Standard errors and goodness-of-fit statistics are reported alongside point estimates.

4.3 Identification and robustness

A low Hurst estimate is not, by itself, sufficient evidence for a structural fractional model. Realized-volatility measurement error, finite-sample effects, jumps, intraday seasonality and microstructure noise can distort short-horizon scaling. The empirical design therefore reports the baseline H estimate together with alternative realized-measure estimates, alternative lag windows, rolling estimates and regime-specific estimates. Statistical uncertainty is quantified with a dependence-preserving block bootstrap. This distinction is essential because H is

© 2026 St. George Capital (SGC). All rights reserved

subsequently carried into option pricing as an empirically identified input rather than selected only to improve calibration fit.

5. Option-Implied Identification: Short-Maturity Skew

Historical time series and option prices encode different probability measures and should not be treated as mechanically interchangeable. Nevertheless, rough stochastic-volatility models generate a sharp qualitative implication for the front end of the implied-volatility surface. Let k=log(K/F_T) denote log-forward moneyness and σ_BS(k,T) the Black-Scholes implied volatility. Define the absolute ATM skew

$$\psi(T) = \left| \frac{\partial \sigma_{BS}(k,T)}{\partial k} \right|_{k=0} . \quad (10)$$

In rough stochastic-volatility asymptotics, the short-maturity skew inherits a power-law dependence on maturity of the generic form ψ(T) ∝ T^{H−1/2}. Thus H<1/2 produces a steepening of the skew as T↓0, whereas the Brownian regularity benchmark does not generate the same singular front-end behaviour. Empirically, ψ(T) will be estimated locally from near-ATM option quotes after transforming strikes into forward log-moneyness and fitting a low-order local polynomial separately by expiry. The resulting term structure provides an option-implied cross-check on the historical roughness estimate rather than a substitute for it.

6. Rough Bergomi Pricing Architecture

The rough Bergomi model provides a parsimonious non-Markovian specification in which the initial forward-variance curve is taken directly from the market and stochastic deviations around that curve are generated by a fractional Volterra factor. Under the risk-neutral measure, a convenient representation is

$$v_t = \xi_0(t) \exp(\eta Y_t - \tfrac{1}{2}\eta^2 t^{2H}) \quad (11)$$

$$\frac{dS_t}{S_t} = r_t\, dt + \sqrt{v_t}\, dZ_t, \qquad dZ_t = \rho\, dW_t + \sqrt{(1-\rho^2)}\, dW_t^{\perp} . \quad (12)$$

Here $\xi_0(t)$ is the initial forward-variance curve, $\eta>0$ controls volatility-of-volatility, $\rho \in [-1,1]$ governs instantaneous spot-volatility dependence, and Y_t is the fractional Volterra process in (5). The compensator $-\tfrac{1}{2}\eta^2 t^{2H}$ ensures that $E[v_t]=\xi_0(t)$ under the chosen normalization. The separation between ξ_0 and the stochastic factor is useful empirically: the observed ATM term structure anchors the variance level, while H, η and ρ govern the roughness, smile curvature and skew.

The baseline calibration will avoid fixing ρ solely by convention. Conditional on an empirically estimated H and a market-implied ξ_0, the remaining parameters will be estimated against a cross-section of liquid option quotes using a transparent loss function in implied-volatility or price space. Monte Carlo error will be controlled through a fixed simulation design and reported separately from calibration error. The principal comparison is not whether rBergomi can interpolate a surface with sufficient flexibility, but whether its structural restrictions improve the representation of short-maturity skew and surface dynamics relative to a conventional Markovian stochastic-volatility benchmark.

7. Quantitative Evidence and Numerical Results

The quantitative stage consists of direct independent raw-data estimation plus controlled numerical diagnostics. First, we estimate the Hurst exponent directly on the S&P 500 realized-volatility panel and not from the supplied scaling summaries. Second, we evaluate the stability of the estimate across realized-volatility measures, lag windows, rolling samples and market regimes, with dependence-preserving bootstrap inference. Third, fractional Brownian paths and the short-maturity skew law are used to isolate the economic significance of the estimated roughness. Fourth, rough Bergomi and Heston surfaces are simulated under comparable starting variance and leverage assumptions to illustrate the pricing mechanism that will be subjected to market calibration when a clean option-quote cross-section is supplied.

The Hurst estimates below are obtained directly from the raw Oxford-Man S&P 500 realized-volatility series. The following option surfaces remain controlled model experiments rather than claims of market prototypes. This distinguishes our independently estimated historical roughness parameter from the subsequent risk-neutral calibration problem.

© 2026 St. George Capital (SGC). All rights reserved

7.1 Raw-data Hurst estimation and robustness

The empirical sample contains 4,189 valid S&P 500 realized-volatility observations from 3 January 2000 to 6 October 2016. The baseline proxy is five-minute realized variance RV_t. We transform it into $x_t = \log(\sqrt{RV_t})$ and estimate q-th order structure functions over lags $\Delta = 1,...,30$ trading days for $q \in \{0.5,1,1.5,2,3\}$. For each q, the first-stage slope ζ_q is obtained from log $m(q,\Delta)$ on log Δ; the second-stage regression of ζ_q on q identifies H.

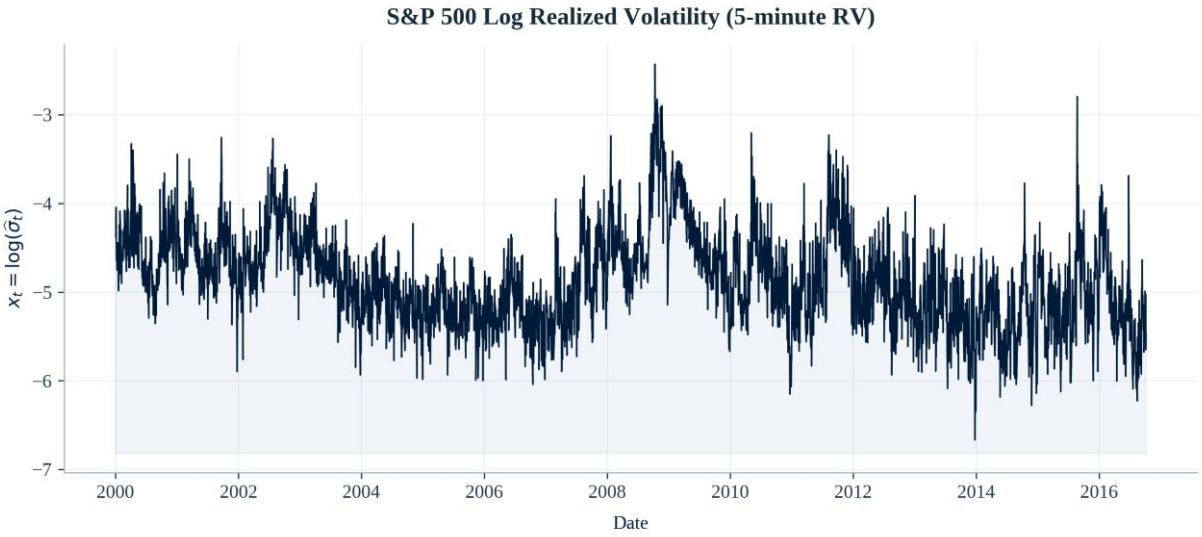


Figure 1. S&P 500 log realized volatility constructed from five-minute realized variance, 3 January 2000-6 October 2016. Source: author computation from Oxford-Man realized measures.

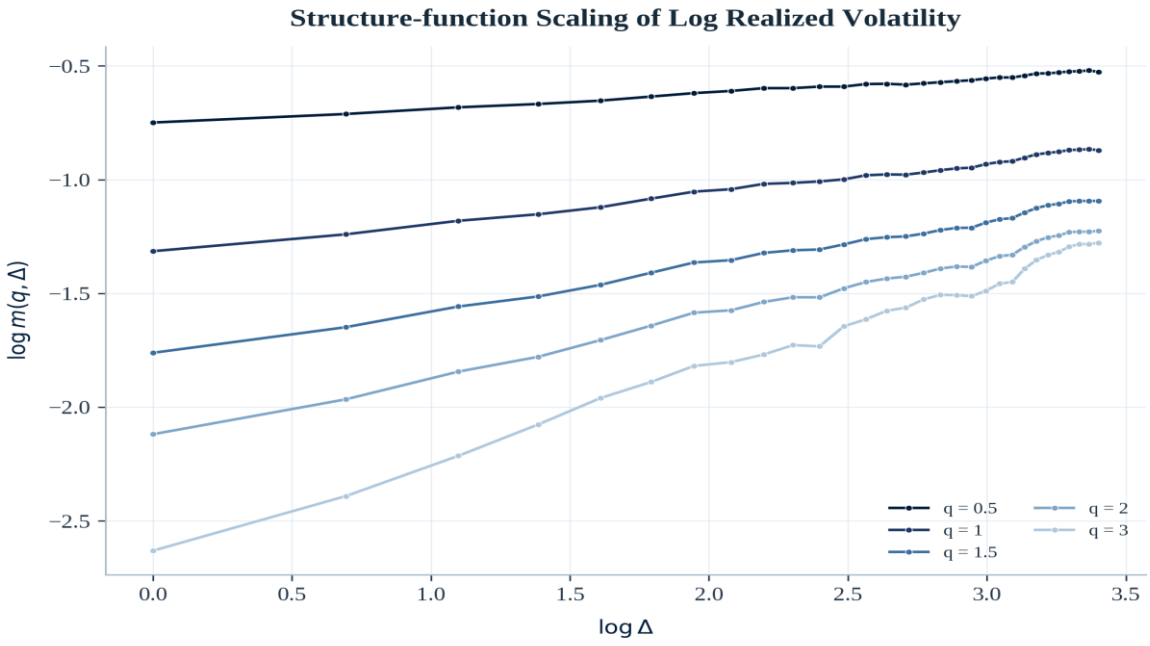


Figure 2. Log-log structure-function scaling across moment orders q and lags 1-30 trading days. Source: author computation.

The baseline second-stage estimate is $\hat{H} = 0.131810$, with intercept 0.002759 and $R^2 = 0.999984$. Imposing the theoretical zero-intercept restriction $\zeta_0 = 0$ gives $\hat{H}_0 = 0.133147$. The near equality of the unrestricted and zero-intercept estimates indicates that the low-H intercept does not drive the low-H result.

© 2026 St. George Capital (SGC). All rights reserved

Quantity	Estimate	Interpretation
$\hat{H}$	0.131810	Baseline slope, five-minute RV, lags 1-30
Intercept	0.002759	Second-stage intercept
R^2	0.999984	Cross-moment linearity
$\hat{H}_0$	0.133147	Slope with intercept constrained to zero
N	4,189	Valid daily realized-volatility observations

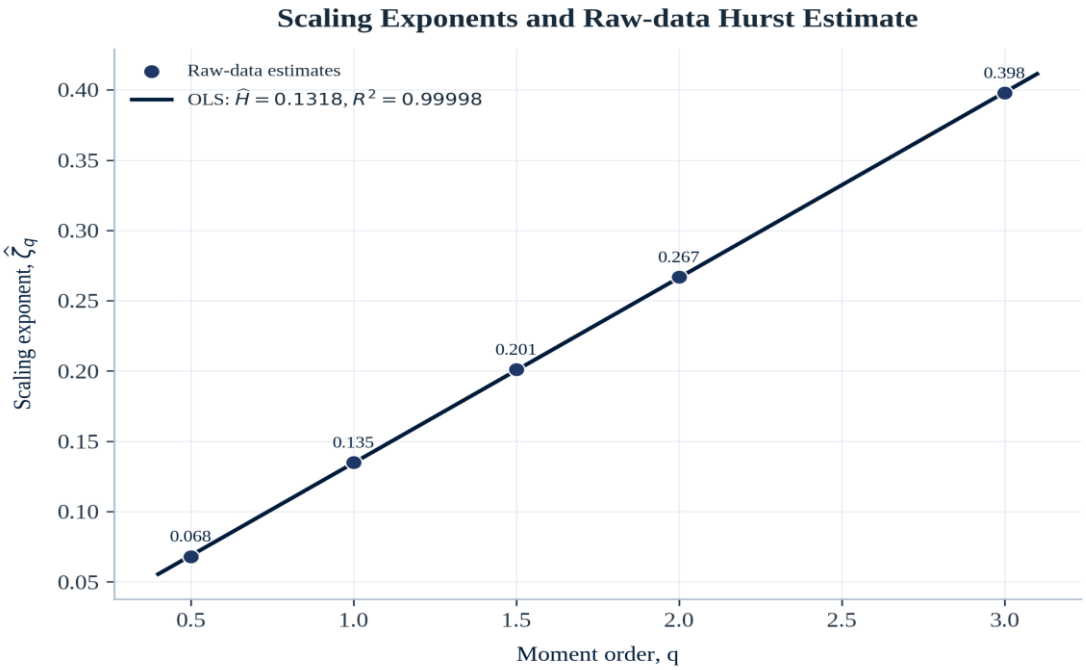


Figure 3. Raw-data scaling exponents and fitted Hurst regression. The slope $\hat{H} = 0.1318$ is estimated directly from the S&P 500 realized-volatility panel.

Statistical uncertainty is assessed by resampling blocks of the joint structure-function contribution vector rather than concatenating blocks of log-volatility levels. This preserves serial dependence and cross-lag/cross-moment dependence while avoiding artificial jumps at block boundaries. With 1,000 replications and 20-day blocks, the percentile interval is [0.107746, 0.154500], with bootstrap median 0.131030.

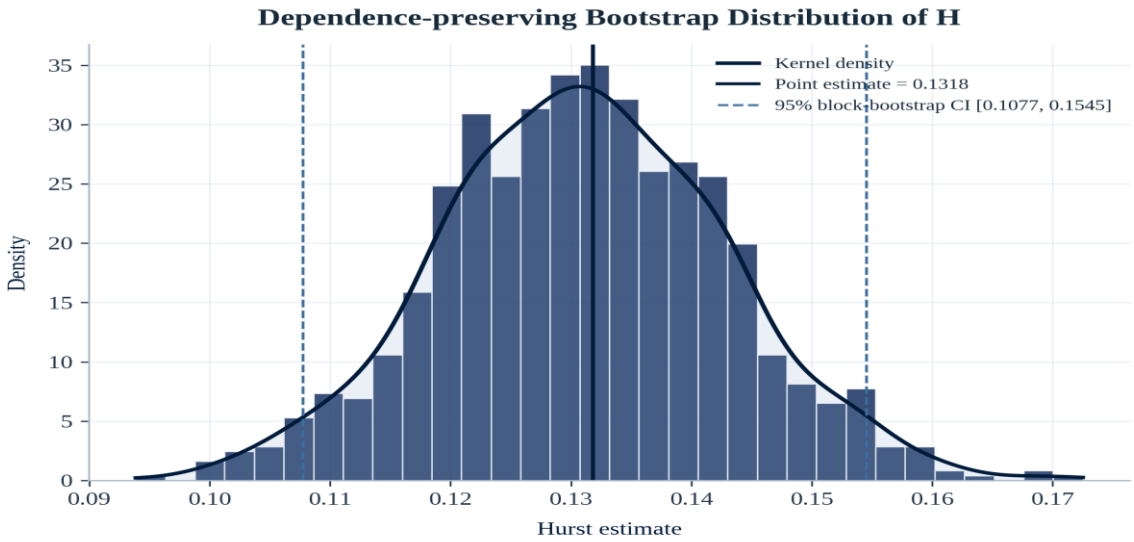


Figure 4. Dependence-preserving moment-block bootstrap distribution of the Hurst estimate (1,000 replications; 20-day blocks).

© 2026 St. George Capital (SGC). All rights reserved

The result is robust to the realized-volatility definition. At the common 1-30 day lag window, $\hat{H}$ equals 0.1599 for subsampled five-minute realized variance, 0.1208 for ten-minute realized variance and 0.1474 for the realized kernel. Across all four measures and lag windows 1-20, 1-30 and 1-60, every estimate remains far below 0.5. The range nevertheless confirms that measurement choice affects the numerical magnitude of roughness and should not be treated as innocuous.

Measure	$\hat{H}$ (1-20)	$\hat{H}$ (1-30)	$\hat{H}$ (1-60)
5-minute RV	0.1301	0.1318	0.1204
5-minute subsampled RV	0.1615	0.1599	0.1397
10-minute RV	0.1167	0.1208	0.1134
Realized kernel	0.1478	0.1474	0.1325

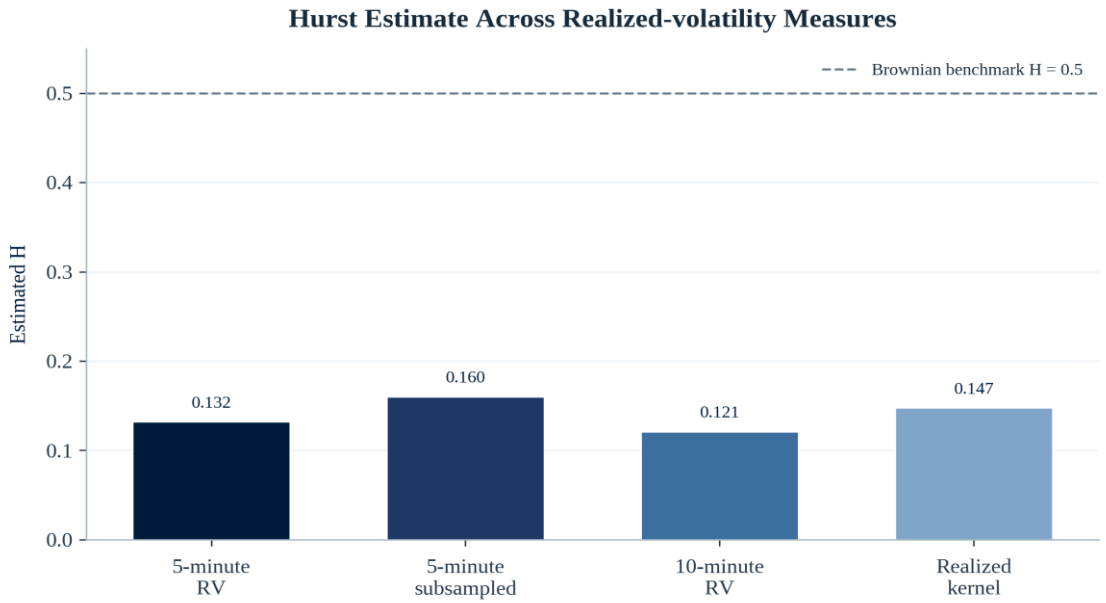


Figure 5. Hurst estimates across alternative realized-volatility measures at the 1-30 day lag window. The dashed line marks the Brownian benchmark H = 0.5.

Roughness is also time-varying. Descriptive regime estimates are 0.1069 for 2000-03, 0.0728 for 2004-06, 0.1858 for 2007-09, 0.1198 for 2010-12 and 0.1343 for 2013-15. The financial-crisis window therefore displays a materially larger H than the pre-crisis 2004-06 window, although both remain in the rough regime. These estimates do not establish structural breaks by themselves; they motivate treating H as an empirically uncertain and potentially regime-sensitive quantity.

Sample Period	N	$\hat{H}$	R^2
2000-2003	985	0.1069	0.9977
2004-2006	752	0.0728	0.9986
2007-2009	749	0.1858	1.0000
2010-2012	754	0.1198	1.0000
2013-2015	756	0.1343	0.9997

© 2026 St. George Capital (SGC). All rights reserved

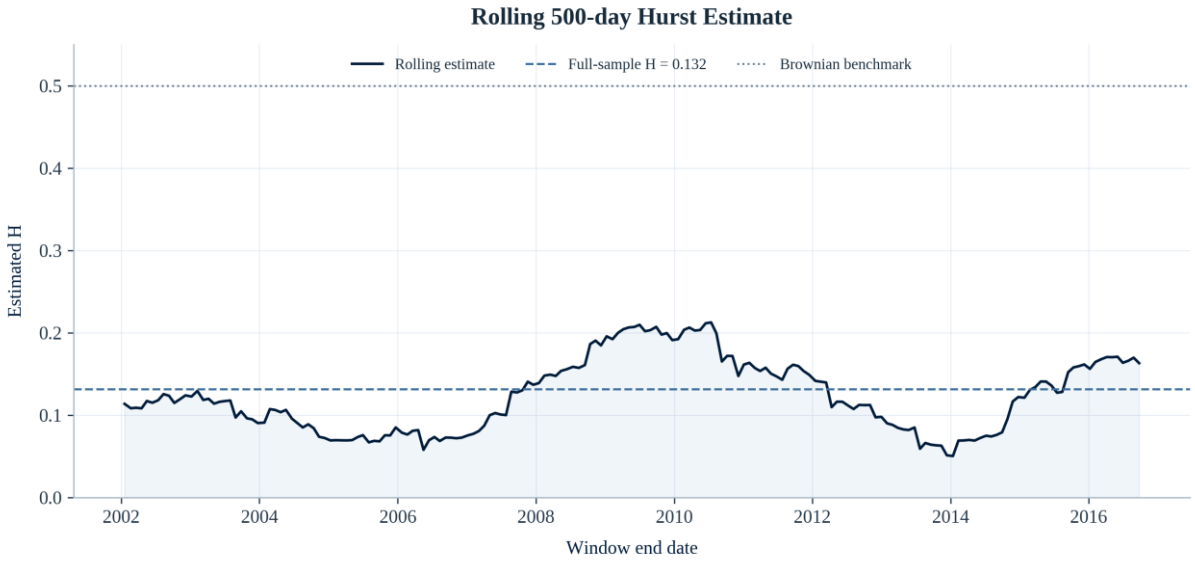


Figure 6. Rolling 500-trading-day Hurst estimate, sampled every 20 trading days. The full-sample estimate and Brownian benchmark are shown for reference.

Taken together, the raw-data results support H well below 1/2 but reject the stronger interpretation that a single numerical value of H is invariant to measurement and market conditions. The empirical contribution is therefore the combination of a stable roughness classification with economically meaningful variation in the estimated degree of roughness.

7.2 Hurst index and path regularity: simulation validation

To isolate the role of H independently of market data, fractional Brownian paths are simulated from the exact Gaussian covariance matrix. The three panels use H = 0.1, 0.5 and 0.9 under the same normalized time horizon. The H = 0.1 trajectory exhibits rapid local oscillation; H = 0.5 recovers Brownian regularity; H = 0.9 generates highly persistent, smooth-looking increments. The experiment is not a calibration result: it is a numerical verification of the mapping between the Hurst index, increment dependence and local Hölder regularity.

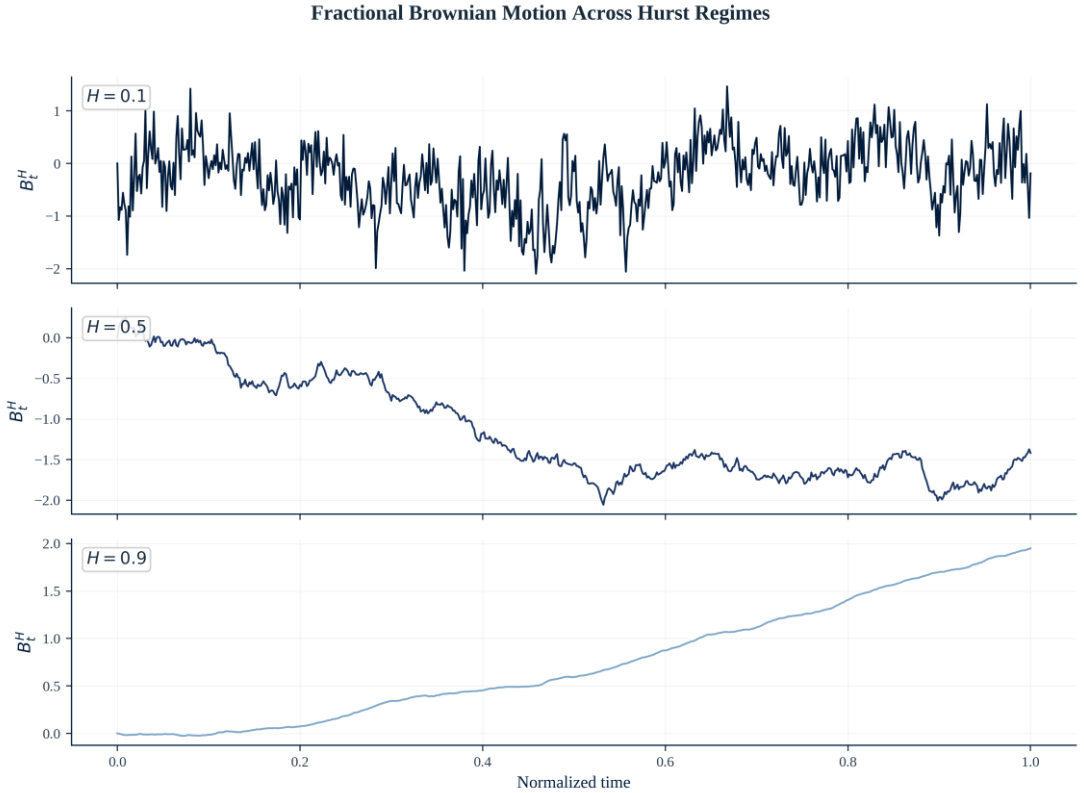


Figure 7. Fractional Brownian motion sample paths for H = 0.1, 0.5 and 0.9. Source: author simulation; fixed random seed for reproducibility.

© 2026 St. George Capital (SGC). All rights reserved

7.3 Short-maturity skew implied by the raw-data roughness estimate

The short-maturity implication can be seen directly from ψ(T) ∝ T^{H−1/2}. The curves below are normalized at a common multiplicative constant, so only the maturity exponent is compared. For H = 0.10, the exponent equals −0.40 and the skew magnitude rises sharply as T approaches zero. For H = 0.50 the exponent is zero, leaving the normalized skew flat. The H = 0.135 curve is included as an alternative empirical benchmark motivated by the robustness literature discussed above.

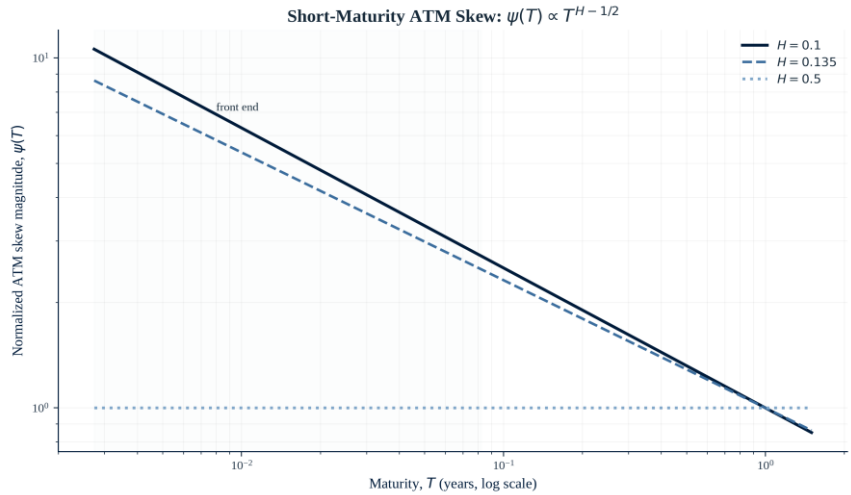


Figure 8. Theoretical ATM-skew power law under alternative Hurst indices. Curves are normalized and illustrate the maturity exponent rather than a market calibration.

7.4 Rough Bergomi numerical experiment

The next experiment embeds the rough factor into a Riemann–Liouville rough Bergomi specification. The variance factor is discretized on a regular grid and the singular kernel is integrated over each time cell to preserve adaptedness. Option prices are computed with conditional Monte Carlo: conditional on the Brownian path driving variance, the orthogonal spot innovation is integrated analytically through a Black–Scholes conditional expectation. This variance-reduction device materially stabilizes the implied-volatility surface at short maturities.

Model	Parameters used in the numerical experiment
Rough Bergomi	H = 0.132; η = 1.35; ρ = −0.75; ξ_0 = 0.04
Heston	$v_0 = \theta$ = 0.04; κ = 2.5; ξ = 0.50; ρ = −0.75
Common setup	S_0 = 1; r = 0; maturities 0.03–0.50 years; identical starting variance 4%

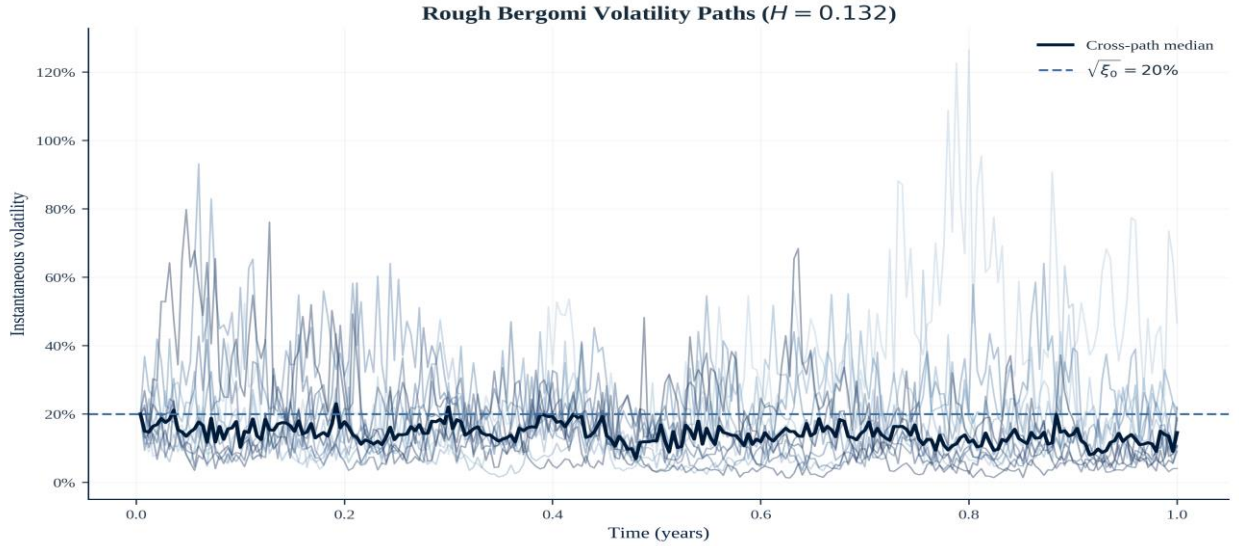


Figure 9. Illustrative rough Bergomi instantaneous-volatility paths. The 20% line corresponds to $\sqrt{\xi_0}$. These paths are simulated, not estimated from market data.

© 2026 St. George Capital (SGC). All rights reserved

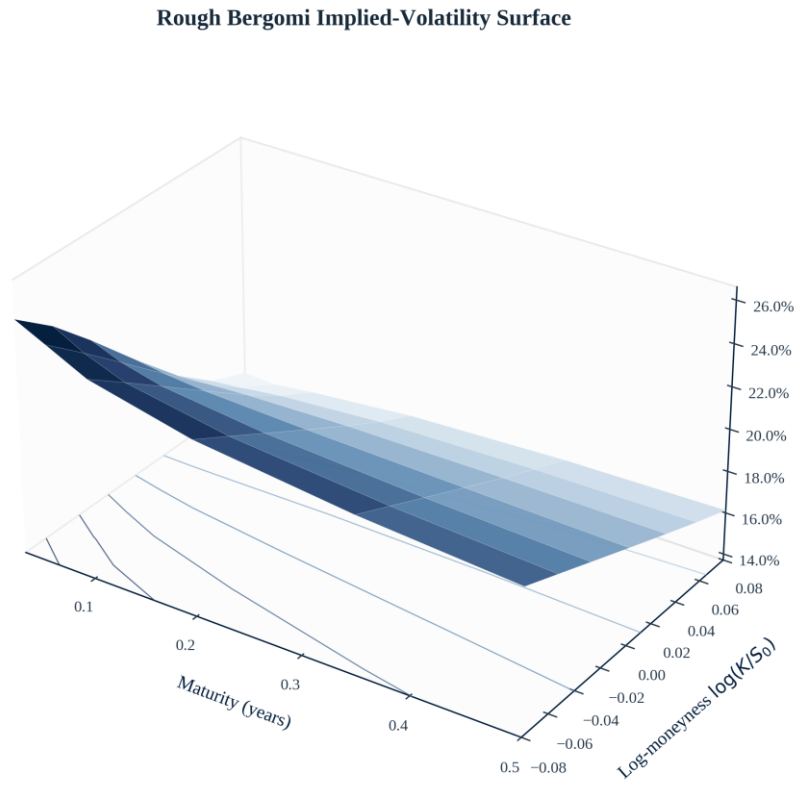


Figure 10. Illustrative rough Bergomi implied-volatility surface generated by conditional Monte Carlo.

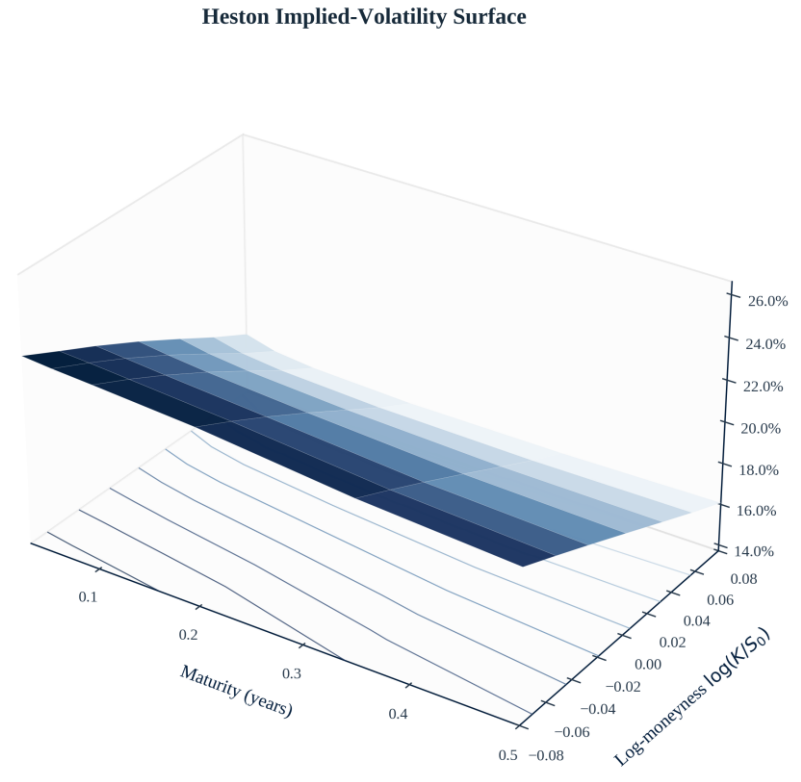


Figure 11. Illustrative Heston implied-volatility surface under a comparable starting variance and leverage parameter.

© 2026 St. George Capital (SGC). All rights reserved

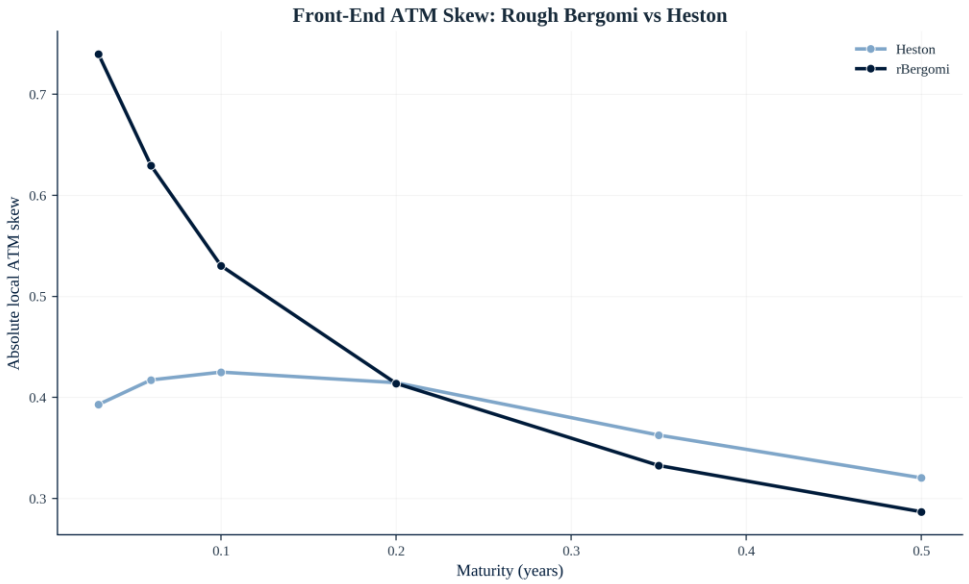


Figure 12. Local ATM-skew magnitude extracted from the simulated surfaces. The rough model displays materially stronger front-end maturity dependence under the stated parameterization.

Under this controlled parameterization, the rough Bergomi surface generates a pronounced short-dated downside skew that decays with maturity, whereas the Heston benchmark produces a much flatter front-end term structure. For example, the absolute local ATM slope in the rough simulation declines from approximately 0.75 at T = 0.03 years to 0.29 at T = 0.50 years; the corresponding Heston values are approximately 0.38 and 0.32. These numbers demonstrate the structural mechanism predicted by rough-volatility asymptotics. They do not establish superior market fit because the parameters were chosen for a controlled comparison rather than calibrated to the same observed option surface.

8. Interpretation, Limitations and Empirical Extension

Four conclusions emerge. First, direct estimation from 4,189 S&P 500 realized-volatility observations yields $\hat{H} = 0.1318$, with a dependence-preserving bootstrap interval [0.1077, 0.1545]. Second, the classification $H < 1/2$ survives alternative realized measures and lag windows, although the point estimate ranges materially across measurement choices. Third, rolling and regime estimates show that the degree of roughness is not numerically constant through time. Fourth, the numerical rough-Bergomi experiment translates low-H regularity into a stronger short-end skew mechanism than the Markovian benchmark under controlled parameters.

The central limitation has shifted from historical-volatility identification to risk-neutral validation. The Hurst estimate is now computed independently from the raw realized-volatility panel, but the implied-volatility surfaces remain model-generated rather than calibrated to a common historical SPX option cross-section. A definitive pricing comparison therefore requires dated option quotes, forward and discount inputs, liquidity filters, joint calibration under a pre-specified loss function, and held-out evaluation. No claim of superior market fit is made before that evidence is available.

A further methodological caveat concerns the interpretation of roughness itself. Recent work by Cont and Das (2024) shows that realized-volatility estimation error can generate apparent $H < 1/2$ even when the latent volatility is not rough. Consequently, the noise-robust realized-measure checks reported above are important for assessing the sensitivity of the low-H result, while the structure-function estimate should not be interpreted as error-free.

9. Conclusion

The empirical evidence supports a robust rough-volatility classification for S&P 500 realized volatility. The baseline raw-data estimate $\hat{H} = 0.1318$ is far below the Brownian benchmark and remains so under subsampled, lower-frequency and realized-kernel measures. At the same time, measurement and regime dependence caution against treating H as an immutable constant. Numerical experiments show how this empirically identified low regularity maps into rough-Bergomi path behaviour and a maturity-sensitive front-end skew. The remaining empirical task is risk-neutral: calibrating rough Bergomi and Heston to the same observed option surface and evaluating their errors on held-out quotes.

© 2026 St. George Capital (SGC). All rights reserved

Appendix A. Computational Implementation and Reproducibility

All empirical estimation, bootstrap inference and figures are generated in Python from the archived S&P 500 realized-volatility CSV. The raw-data pipeline (i) cleans positive realized-variance observations, (ii) constructs log realized volatility, (iii) computes structure functions jointly across q and Δ, (iv) estimates first- and second-stage scaling regressions, (v) implements a moment-block bootstrap on the joint contribution matrix, and (vi) produces measure, lag-window, rolling and regime robustness diagnostics. Fixed random seeds are used for bootstrap and simulation routines. The accompanying code and CSV outputs reproduce every numerical result reported in Section 7.1.

The option-pricing code is retained separately from the raw-data estimator so that historical roughness identification is not conflated with risk-neutral calibration. The controlled rough-Bergomi and Heston figures are reproducible from the accompanying numerical script. A market-calibration module is included as a template but is intentionally not reported as empirical evidence until a clean option-quote panel is supplied.

References

Bayer, C., Friz, P. K., & Gatheral, J. (2016). Pricing under rough volatility. Quantitative Finance, 16(6), 887–904. https://doi.org/10.1080/14697688.2015.1099717

Black, F., & Scholes, M. (1973). The pricing of options and corporate liabilities. Journal of Political Economy, 81(3), 637–654.

Comte, F., & Renault, E. (1998). Long memory in continuous-time stochastic volatility models. Mathematical Finance, 8(4), 291–323.

Fukasawa, M. (2011). Asymptotic analysis for stochastic volatility: martingale expansion. Finance and Stochastics, 15(4), 635–654.

Gatheral, J., Jaisson, T., & Rosenbaum, M. (2018). Volatility is rough. Quantitative Finance, 18(6), 933–949. https://doi.org/10.1080/14697688.2017.1393551

Heber, G., Lunde, A., Shephard, N., & Sheppard, K. (2009). Oxford-Man Institute's Realized Library, version 0.3. Oxford-Man Institute, University of Oxford.

Heston, S. L. (1993). A closed-form solution for options with stochastic volatility with applications to bond and currency options. Review of Financial Studies, 6(2), 327–343.

Cont, R., & Das, P. (2024). Rough volatility: Fact or artefact? Sankhya B, 86, 191–223. https://doi.org/10.1007/s13571-024-00322-2

© 2026 St. George Capital (SGC). All rights reserved